



On-orbit performance analysis of spinning spacecraft magnetic control laws. Application to the UPMSat-2 mission

Angel Porras-Hermoso ^{a,b,*}, Javier Piqueras ^{a,b}, Javier Cubas ^{a,b}, Elena Roibás-Millán ^{a,b}

^a Universidad Politécnica de Madrid, Pza. del Cardenal Cisneros 3, Madrid, 28040, Spain

^b Instituto Universitario de Microgravedad 'Ignacio Da Riva' (IDR/UPM), Pza. del Cardenal Cisneros 3, Madrid, 28040, Spain

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ABSTRACT

This paper studies the performance of a new magnetic attitude control law developed for the UPMSat-2 satellite. The UPMSat-2 control law, which is a modification of the B-dot, allows the satellite to align one axis with the normal to the orbital plane and achieve a target angular velocity, without requiring attitude determination. The study evaluates the control law's effectiveness using the first housekeeping data received since the satellite's launch in September 2020. To verify the control's on-orbit performance, data from sun sensors and magnetometers, along with thermal analysis of the satellite external temperatures, are used to confirm that the satellite remains at its design attitude and angular velocity. The study concludes that the proposed control law is a simple and effective alternative for small satellite attitude control.

1. Introduction

Attitude control systems have a key role in spacecraft operations since they provide an essential feature to achieve mission goals and even ensure the survival of the spacecraft. There are a high number of approaches to be able to control the attitude dynamics [1], and many more are being developed in recent years with a focus on small-size satellites [2]. In that case, and for satellites operating in Low Earth Orbits (LEO), a particularly effective form of attitude control is based on the measurement and interaction with the Earth's magnetic field. Electromagnetic actuators provide a simple solution able to generate torques onboard the spacecraft by the interaction between three orthogonal magnetic coils and the Earth's magnetic field.

The first satellite using passive magnetic control systems was Transit 1B [3], followed by Tiros II [4] that incorporated an active magnetic control, both launched in 1960. Attitude reconstruction using measurements of the Earth's magnetic field was first performed with Sputnik-3 satellite, launched in 1958 [5]. Despite the widespread use of magnetic control systems, they usually play a secondary role, being almost exclusively oriented to perform specific tasks such as detumbling, reaction wheel desaturation or to provide attitude control in safe modes due to its simplicity and low failure risks. However, there are notable exceptions, such as in spin-stabilized satellites, where magnetic control plays a crucial part [6–8].

In recent years, the use and applications of magnetic control laws have changed. The increasing interest in micro-, nano- and even smaller satellites, whose attitude requirements usually have lower demands on

accuracy, have made magnetic control laws an essential component to meet the weight, power, and low complexity requirements of these missions [5]. Magnetic actuators present very interesting properties such as significant savings in overall weight and complexity of the system, lack of catastrophic failure modes, the possibility to modulate the control torque and very larger operational life. These factors have led to an extensive development of purely magnetic control laws.

Magnetic control systems can be classified into passive and active systems. Passive systems were extremely popular in the first decades of space exploration. These systems included permanent magnets and/or hysteresis rods, which were used to detumble and stabilize the satellite along the local geomagnetic vector. These systems are cheap, simple, reliable, and do not require computation. However, their accuracy and maneuverability are poor and limited. Active magnetic systems are nowadays the preferred systems due to the advance in component miniaturization and computer performance. These systems include magnetorquers, that induce a dipole moment when an electrical current is applied. The dipole moment interacts with the external geomagnetic field, producing the control torque.

One of the most widely used algorithms for detumbling is the so-called B-dot [9]. In its simplest form, the detumbling mode based on the B-dot law is based on negative feedback of the derivative of the measured magnetic field vector, obtained by a magnetometer. This algorithm, although been relatively old, is still used and investigated [10,11]. B-dot is an active purely magnetic law that can achieve

* Corresponding author at: Universidad Politécnica de Madrid, Pza. del Cardenal Cisneros 3, Madrid, 28040, Spain.

E-mail address: angel.porras.hermoso@upm.es (A. Porras-Hermoso).

a reasonably good accuracy and almost stop the rotation of the satellite, as the angular velocity of the satellite tends to shrink around 1.8 times the orbital angular velocity [12]. Regarding its time response, it depends on both the magnetorquer maximum dipole and the orbit inclination, as it has been shown in more recent works [13,14].

The main disadvantage of magnetic controlled systems is their inherent underactuation, as they can only produce a magnetic torque in the perpendicular direction of the Earth's magnetic field, making 3-axis stabilization costly and difficult to achieve. In recent years different attempts to obtain a purely magnetic (by only employing magnetic actuators) 3-axis stabilization control law have been proposed. These algorithms use the periodicity of the Earth's magnetic field to overcome the underactuation problem and achieve 3-axis control along the orbit [15,16]. Nevertheless, most of these methods are theoretical and have not been proven on a real mission. Moreover, these methods usually require Earth's magnetic field models and solving complex optimization problems that could exceed the computational power of the on-board computers in small satellites.

A simpler solution to solve the underactuation problem is to complete the magnetic control with other actuators, as an example, the system can use momentum wheels and magnetorquers to build a bias-momentum attitude control system [17]. Other approaches use the disturbances to stabilize the satellite. For example, the torque produced by the gravity gradient can be used to obtain a 3-axis stabilization in combination with a magnetic control law [18–20]. Other methods involve the use of aerodynamic drag to stabilize the satellite. These methods can only be used in satellites located in LEO and with a proper mass distribution. With the aerodynamic force the satellite can be stabilized in 2 axes, and the magnetic control only needs to actuate in one axis. Good examples of this kind of control laws can be found in the literature [21–23]. Other interesting magnetic control laws are the ones used in spinning spacecrafts [24–28]. These algorithms use the property that spinning around the maximum moment of inertia is stable. However, small errors and external torques can induce nutation motion to the satellite, so it is necessary to counteract these terms with a control torque produced by the magnetorquer. In Ref. [29], a modification of the B-dot law is proposed which provides an effective control of the angular rotation of the satellite, enabling the rotation axis to be aligned with the normal of the orbit without using any reaction wheel.

This paper analyzes and discusses the on-orbit behavior of the magnetic control law on board the UPMSat-2. This magnetic control law (described in Section 2) can be considered an improvement of the B-dot algorithm for spin-stabilized spacecrafts to include the full control of the rotation velocity around its main inertia axis. One of the main advantages of this law is that attitude determination is not required. Therefore, other attitude sensors, apart from magnetometers, are not necessary, eliminating potential errors derived from attitude determination, as the ones happened in Compass-1 mission [30]. Consequently, it can be said that this control law is purely magnetic concerning its actuators and sensors. This law was briefly used in Hessi mission [31] for detumbling purposes, but its use as main control and its effect in the stabilization and orientation of the spacecraft was not analyzed. Those aspects were previously studied theoretically [29] and the results are correlated and verified in this work, using UPMSat-2 flight data.

The UPMSat-2 is a 50 kg-class microsatellite developed by the *Instituto Universitario de Microgravedad “Ignacio Da Riva”* (IDR/UPM),¹ a research institute integrated in the *Universidad Politécnica de Madrid* (UPM), and STRAST² group. The main purpose of this satellite is to serve as a technology demonstrator of components and algorithms designed by IDR/UPM and different partners (Tecnobit, SAFT, Iberespacio, Bartington, etc.). Furthermore, this satellite proved to be a greatly valuable educational tool for students of the Master in Space Systems

(MUSE *Máster Universitario en Sistemas Espaciales*) of UPM. The most relevant characteristics of UPMSat-2 are summarized in Table 1.

This paper is organized as follows. In Section 2, the control law governing the UPMSat-2 attitude is presented and explained. In Section 3, the main problems encountered regarding attitude calculation are discussed, and a method to obtain the angular velocity of the spacecraft is described. The results obtained are presented and discussed in Section 4. Finally, in Section 5, the conclusions extracted from this work are summarized.

2. UPMSat-2 attitude control

The dynamic model of a rigid satellite in the satellite's body reference frame, indicated with the superscript s (F^s), can be expressed using the Euler equation [32]:

$$\mathbf{I}\dot{\boldsymbol{\omega}}^s = -\boldsymbol{\omega}^s \times \mathbf{I}\boldsymbol{\omega}^s + \mathbf{T}_{coil}^s + \mathbf{T}_{dist}^s \quad (1)$$

where $\mathbf{I}$ is the inertia matrix of the satellite in the body reference frame, $\boldsymbol{\omega}^s$ is the angular velocity vector of the satellite relative to the inertial reference frame, $\mathbf{T}_{coil}^s$ is the control magnetic torque, and $\mathbf{T}_{dist}^s$ is the torque produced by the environment disturbances, both in the body frame.

The UPMSat-2 is almost axisymmetric around its z^s -axis, so its inertial matrix in its principal body axes (principal axes of inertia) can be approximated as:

$$\mathbf{I} = \begin{bmatrix} I_{\perp} & 0 & 0 \\ 0 & I_{\perp} & 0 \\ 0 & 0 & I_{\parallel} \end{bmatrix} \quad (2)$$

The control torque is the result of the interaction between the magnetic dipole moment of the magnetorquers, $\mathbf{m}$, and the geomagnetic field $\mathbf{B}$, and it can be expressed as follows:

$$\mathbf{T}_{coil}^s(t) = \mathbf{m}(t) \times \mathbf{B}(t) \quad (3)$$

In only magnetic controlled system, the magnetic moment generated by the magnetorquers is the only controllable actuator of the system. To regulate this magnetic moment, the following control law is proposed [33]:

$$\mathbf{m}^s = -k(\dot{\mathbf{B}}^s + \boldsymbol{\omega}_d^s \times \mathbf{B}^s) \quad (4)$$

where k is the control gain, $\mathbf{B}^s$ is the Earth's magnetic field in the body frame, and $\boldsymbol{\omega}_d^s$ is the desired angular rate. In the case of UPMSat-2, the commanded angular speed of the satellite is parallel to the axisymmetric principal axis, which corresponds to the axis of minimum inertia. However, it is important to note that the desired angular velocity cannot be set arbitrarily, as its magnitude must satisfy the following condition:

$$\omega_d \gg \Omega_o \quad (5)$$

where Ω_o is the orbital rotation rate.

Therefore, the control torque can be expressed as:

$$\mathbf{T}_{coil}^s(t) = -k(\dot{\mathbf{B}}^s + \boldsymbol{\omega}_d^s \times \mathbf{B}^s) \times \mathbf{B}^s \quad (6)$$

By introducing Eq. (6) in (1), and neglecting the effects of the disturbance torque, the satellite dynamic equation can be written as:

$$\mathbf{I}\dot{\boldsymbol{\omega}}^s = -\boldsymbol{\omega}^s \times \mathbf{I}\boldsymbol{\omega}^s - k(\dot{\mathbf{B}}^s + \boldsymbol{\omega}_d^s \times \mathbf{B}^s) \times \mathbf{B}^s \quad (7)$$

As it is known, the derivative of a quantity, such as the magnetic field, with regard to moving axes, such as those of the satellite, can be expressed in the following manner:

$$\dot{\mathbf{B}}^s = \mathbf{A}_1^s \dot{\mathbf{B}}^1 + \mathbf{B}^s \times \boldsymbol{\omega}^s \quad (8)$$

¹ <https://www.idr.upm.es/>

² <https://www.dit.upm.es/~str/>.

Table 1
UPMSat-2 main characteristics.

Orbit	• Sun-synchronous orbit. • Local Time of Descending Node: 10:30 AM. • Altitude at launch: 518 km.
Dimensions	0.5 m × 0.5 m × 0.6 m.
Mass and Moment of inertia	Mass: 50 kg. Moment of inertia, z-axis: 1.833 kg m ² Moment of inertia, x-axis: 2.543 kg m ² Moment of inertia, y-axis: 2.525 kg m ²
Attitude control	Magnetic control: • 3 magnetometers (SSBV and Bartintong). • 3 Magnetorquers (ZARM Technik AG) . • Purely magnetic control law designed by IDR/UPM. Expected performance: • Satellite's z-axis oriented perpendicular to the orbital plane. • Rotation about z-axis at an angular velocity of 0.1 rad/s.
Attitude determination	Attitude determination on ground with received telemetry. Sensors: • Magnetometers. • Solar panels. • Solar sensors designed by IDR/UPM.
Thermal control	Passive for all the satellite excluding the battery (thermostats and heaters).
Power	Based on solar photovoltaic panels and batteries: • 5 body-mounted solar panels (Selex Galileo SPVS-5 module with Azur Space 3G28C triple junction solar cells). • 18 A h Li-ion battery designed by Saft. • Direct Energy transfer (DET).
On board Electronic Box	FPGA based (designed by Tecnobit S.L and programmed by STRAST/UPM). Includes: • On-board computer. • Data Handling. • Power supply control. • Power supply distribution.
Communications	• Half-duplex UHF at 400 MHz and 437 MHz. • Baud rate: 1200 Bd (2-FSK) and 9600 Bd (GMSK). • 4 monopole antenna system with circular polarization. • Ground station located in Madrid, Spain. Software programmed by STRAST; hardware configuration supervised by INTA^a
Payload	Technological demonstrators: • Magnetometer (Bartington). • Electronic Box (Tecnobit). • Rotation wheel (SSBV). • Thermal microswitch (Iberespacio). • Battery life (Saft). • Solar sensors (IDR/UPM). • Thermal control (IDR/UPM). Science: • Earth Magnetic Field.

^a INTA: Instituto Nacional de Técnica Aeroespacial, Madrid, Spain.

where A_1^s is the direct cosine matrix between the inertial reference frame and the moving reference frame. By substituting the aforementioned expression into Eq. (7), the following equation is derived:

$$\dot{\mathbf{I}}\boldsymbol{\omega}^s = -k\mathbf{B}^s \times (\boldsymbol{\omega}^s - \boldsymbol{\omega}_d^s) \times \mathbf{B}^s - kA_1^s(\dot{\mathbf{B}}^1 \times \mathbf{B}^1) - \boldsymbol{\omega}^s \times \mathbf{I}\boldsymbol{\omega}^s \quad (9)$$

During the correction phase, when the angular velocity of the satellite is significant, and given an arbitrary value of the satellite's rotation rate different for the target one, it is reasonable to assume the following condition:

$$\|\boldsymbol{\omega} - \boldsymbol{\omega}_d\| \gg \Omega_o \quad (10)$$

As a consequence of condition presented in Eq. (10), the order of magnitude of the second term on the right side of Eq. (9) is negligible compared with the first term. This is due to the variation of the Earth's magnetic field in the inertial reference frame as it changes at an approximate rate of $2\Omega_o$. The effect of Earth's rotation on the magnetic field can be considered negligible since the change in the direction of the magnetic field due to the position of the satellite ($2.16 \cdot 10^{-3}$ rad/s), is much greater than the rotation of the Earth ($7.27 \cdot 10^{-5}$ rad/s). Hence, Eq. (9) for the correction phase can be simplified as shows Eq. (11):

$$\dot{\mathbf{I}}\boldsymbol{\omega}^s = -k\mathbf{B}^s \times (\boldsymbol{\omega}^s - \boldsymbol{\omega}_d^s) \times \mathbf{B}^s - \boldsymbol{\omega}^s \times \mathbf{I}\boldsymbol{\omega}^s \quad (11)$$

Eq. (11) is similar to the one obtained for the B-dot attitude control and, as it was proven in [29], it is asymptotically stable to the desired

rotation rate using the Lyapunov stability criteria. The selection of the value of parameter ω_d^s significantly influences the dynamics of the problem. A larger discrepancy between the target angular velocity and the initial value leads to a more rapid convergence of the control towards the desired value and, therefore, the sooner the stabilization phase will be reached.

Once the angular rotation rate of the satellite approaches to the target one, that is:

$$\|\boldsymbol{\omega} - \boldsymbol{\omega}_d\| \sim \Omega_o \quad (12)$$

the second term of Eq. (9) is no longer negligible, and it produces the following torque:

$$\boldsymbol{\tau} = -kA_1^s[\dot{\mathbf{B}}^1 \times \mathbf{B}^1] \quad (13)$$

It has been previously proved that the satellite has a tendency to align itself with the direction of the torque, $\boldsymbol{\tau}$ [29]. Therefore, a stabilization time will come when it is assumed that the satellite's z-axis is aligned with the torque direction. In this case, for the stabilization phase, Eq. (9) can be rewritten as follows:

$$\dot{\mathbf{I}}\boldsymbol{\omega}^s = -k\mathbf{B}^s \times (\boldsymbol{\omega}^s - \boldsymbol{\omega}_d^s) \times \mathbf{B}^s - k\|\dot{\mathbf{B}}^1(t) \times \mathbf{B}^1(t)\|\mathbf{e}^s - \boldsymbol{\omega}^s \times \mathbf{I}\boldsymbol{\omega}^s \quad (14)$$

where e^s is the unit vector, which has the same direction as the torque, τ and can be calculated as follows:

$$e = \left(\frac{\dot{B}_1^s(t) \times B_1^s(t)}{\|\dot{B}_1^s(t) \times B_1^s(t)\|} \right) \quad (15)$$

Introducing the following change of variable:

$$\omega_d^{*s} = \left(\omega_d + \frac{\|\dot{B}_1^s(t) \times B_1^s(t)\|}{\|B(t)\|^2} \right) e^s \quad (16)$$

and substituting it into Eq. (14), it becomes evident that the resulting equation bears a strong resemblance to the one derived in Eq. (11)

$$I\dot{\omega}^s = -k B^s \times (\omega^s - \omega_d^{*s}) \times B^s - \omega^s \times I \omega^s \quad (17)$$

This new variable, ω_d^{*s} , represents the equilibrium angular velocity that varies periodically with time, t , as the satellite moves along its orbit. Consequently, the steady solution of Eq. (17) is influenced by the control gain, k . For large values of k , the equilibrium angular velocity approaches the exact value of ω_d^{*s} and exhibits a periodic oscillation around the commanded angular velocity between $\omega_d^{*s} + 1.5\Omega_o$ and $\omega_d^{*s} - 3\Omega_o$. For small values of k , the variation of that oscillations is less pronounced, tending ω_d^s to an average value of ω_d^{*s} , approximately $\omega_d^{*s} + 1.7\Omega_o$. Although the average value is slightly higher than ω_d^s this can be easily corrected by commanding a value of ω_d^s that accounts for that increase. However, the oscillations around the equilibrium angular velocity cannot be completely eliminated. Nevertheless, they are a minor compromise given the simplicity of the control. Another factor to take into account at the stabilization stage is the value of the parameter ω_d^s . For high values of ω_d^s ($\|\omega_d^s\| \gg \Omega_o$) the oscillations that occur will be of little importance compared to the value of the target angular velocity. Conversely, if this value is low, the oscillations may approach magnitudes comparable to the target angular velocity, potentially jeopardizing the stability, particularly at lower speeds. Further details on the theoretical performance of this control law can be found in the work of Ref. [29].

The implementation of this algorithm in UPMSat-2 satellite can be found in reference [34] and the main characteristics of the hardware and control parameters are shown in Table 2. The selection of $\omega_d = [0, 0, 0.1]$ rad/s is due to the fact that a high value of the target angular velocity compared to the initial one accelerates the detumbling phase. Once the satellite is stabilized, and after commissioning, subsequent adjustments of the angular velocity via telecommand are possible. Based on thermal and power studies [35], it is apparent that the rotation can be lowered to $5.52 \cdot 10^{-3}$ rad/s and the satellite will remain operationally. However, it is important to note that this value closely approaches the theoretical lower limit, and it is not recommended to go below it. Regarding the upper limit of the target angular velocity, it is given by the sampling rate of the magnetometers, being necessary that the frequency of change of the measured magnetic field be at most half the sampling rate of the magnetometers. Therefore, the maximum rotation frequency of the satellite will be half the sampling rate of the magnetometers if only two consecutive measurements are used to obtain the derivative of the magnetic field ($f_{rot} < \frac{f_{sampling}}{2}$).

One of the primary challenges in implementing this algorithm in real-world scenarios is the unreliability of magnetometer measurements while magnetorquers are active. The magnetometer's readings are influenced by the magnetorquers' magnetic field, making it necessary to ensure that the magnetometer reading and the magnetorquer activation occurs at different time instants. To accomplish this, a control cycle was devised (see Fig. 1), with a duration of 2 s, divided into two separate intervals of one second each. During the first interval, the magnetometers are sampled with a period of 200 ms. These measurements are averaged, and the derivative of the magnetic field is calculated. By adopting this approach, the measured magnetic field derivative becomes more robust to the influence of noise and disturbances. Then, these results are fed to the control algorithm, which

Table 2

Characteristics of the of selected hardware and parameters for the attitude control of the UPMSat-2.

Characteristic	Description
Magnetorquer	Linear dipole range: $\pm 15 \text{ Am}^2$ Linearity error $< 2\%$ at max. linear moment Residual dipole $\leq 0.5\%$ of max. linear moment
Magnetometer (from SSBV)	Error $< 25 \text{ nT}$ Noise $< 2 \text{ nT (RMS)}$
Control parameters	Control Gain: $k = 1 \cdot 10^8$ Target rotation rate: $\omega_d = [0, 0, 0.1] \text{ rad/s}$

takes a negligible time to process the control action. In the second interval, the control action is transmitted to the magnetorquers, which may take approximately 200–500 ms to complete. The time reserved for this action is influenced by the time needed for the magnetorquer to activate and reach the demanded magnetic moment value, which is usually around 150 to 200 ms. To this time is added another period for the magnetorquer to remain active, which ultimately culminates in a maximum total time of 500 ms. Finally, the control cycle is concluded with a rest period of at least 500 ms. This rest period prevents any residual magnetic moment in the magnetorquers from interfering with the magnetometers' readings.

In Ref. [29], different simulations were done for a satellite with similar characteristics to the UPMSat-2. These simulations consisted in a Montecarlo analysis of the attitude control's behavior once it was stabilized, for different orbit inclination, with constant control gain, k , for each Montecarlo analysis. In Fig. 2 the results for the control gain implemented in the UPMSat-2 are shown. For this case, the results of interest correspond to a inclination of 97.4 deg. For this inclination, a mean deviation around 2.5 deg from the normal direction of the orbit is expected. In terms of the stabilized rotation rate, the deviation from the target rotation rate ranges from 1.25 to 3 times the orbital rotation rate. For the special case of the target rotation rate being 0.1 rad/s, the expected response and values once the satellite is stabilized are shown in Fig. 3, giving as a result a rotation rate of 0.1 rad/s plus 1.7 Ω_o on the z-axis, with small periodical variations of less than Ω_o .

3. UPMSat-2 attitude determination

3.1. The problem of determining UPMSat-2 attitude

UPMSat-2 data is transmitted to the ground station when the satellite has direct access to it. The satellite typically passes over the ground station four times per day (twice during daylight and twice at night). During transmission windows that last around 10–15 min, the satellite's housekeeping information is sent every minute for further analysis. With this information, the attitude of the satellite can be calculated using two independent sensor measurements, and its corresponding modeled magnitudes in the reference frame. The result of this procedure is an "image" of the satellite's orientation at that instant, which is independent of the data rate. The estimation of the angular velocity requires consecutive measurements and the sample rate shall be equal to or greater than twice the highest frequency of the movement, as stated by the Nyquist-Shannon sampling theorem [36]. Since UPMSat-2 does not have rate sensors, the estimation of angular velocity depends on the data rate. It is true that inside the satellite, the sampling rate of the magnetometers allows to calculate the derivative of the magnetic field, which in turn will allow to calculate the angular velocity. However, these data are neither stored nor transmitted. The only information that is available from the UPMSat-2 is that which is sent by its commissioning mode. In its commissioning mode the satellite only sends the current state of the satellite every minute, which does not include the angular velocity of the satellite. This means that the received sample rate of the measurements of the satellite is very close to

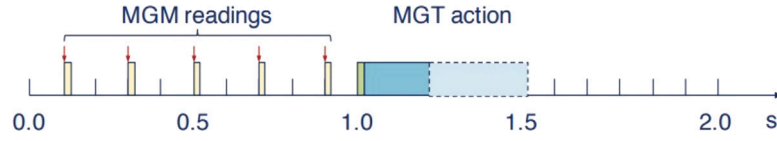


Fig. 1. Control cycle timing.

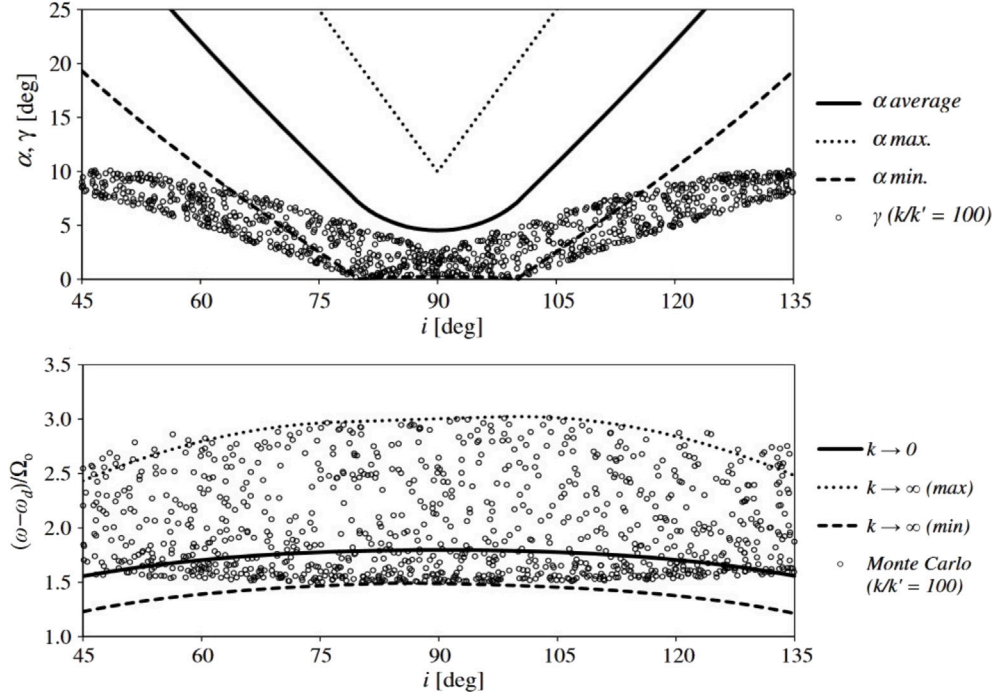


Fig. 2. Results of Monte Carlo analysis, carried out to study the situation once the rotation rate is stabilized. Deviation of the pointing direction from the normal of the orbit (Up). Non-dimensional deviation of the stabilized rotation rate against the target rotation rate (Down). ($k' = 1 \cdot 10^6$ in the graphs' legends) [29].

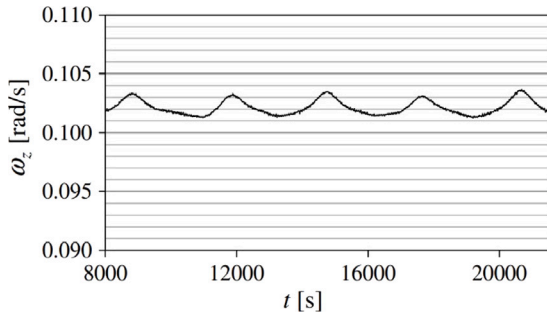


Fig. 3. Example of the expected oscillations in z-axis angular velocity after stabilization [29].

the theoretical rotation rate of the satellite. As a result, it is impossible to directly estimate the spacecraft rotation velocity with consecutive measurements. Additionally, only 10–12 data packages are received in each pass above the ground station, making it difficult to follow the satellite state between passes.

In the available literature, several works related to the analysis of the attitude control subsystem have been published. In most of them in-flight data are analyzed to study the performance of the attitude determination and control subsystems (ADCS) [30,37]. However, the methods used in these papers are not applicable for the UPMSat-2 case, as the satellite does not include gyroscopes or optical devices. Other satellites, such as Tian Tuo 1 [38], use an attitude determination subsystem composed of solar sensors and magnetometers. In

some cases, the attitude can be computed using magnetometers and solar panels [39]. This approach is interesting for the UPMSat-2 case, as its solar sensors are part of the payload and the verification of their in-flight performance depends on the magnetometer results [40]. However, this method can only estimate the spacecraft's orientation and not its angular velocity, which is necessary to prove the main point of the UPMSat-2 attitude control. The work of Ref. [41] proposes a magnetometer-only attitude determination algorithm. With this method, the algorithm can estimate both the orientation and the angular velocity using the information of three consecutive measurements from the magnetometer. However, considering the low data rate received from UPMSat-2 (one data package per minute) and its expected angular velocity (one revolution per minute), this method is not suitable for the UPMSat-2 case. Other methods to determine the attitude of small spinning spacecraft using only magnetometers have been proposed [42], but the apparent angular velocity calculated with these algorithms could differ from the true angular velocity of the spacecraft.

Alternatively, more advanced methods could be considered, which combine the knowledge of the vehicle's dynamics and the sensor readings to estimate the full state of the satellite. This is the case of algorithms like the Kalman filter and its variations. One of the most popular applications in small satellites is to combine measurements from magnetometers and solar sensors into a Multiplicative Extended Kalman Filter (MEKF) [43]. In other algorithms, the spacecraft's attitude is first determined using single value decomposition (SVD), and the result is integrated into an Extended Kalman Filter (EKF) to estimate the angular velocity [44]. More advanced algorithms can estimate the attitude and angular velocity using a magnetic-only algorithm with a

two-step EKF [45]. Once again, the low data rate of measurements from the satellite makes estimating the angular velocity challenging, even for Kalman filters. This is due the filter may find it difficult to follow the true state, as the dynamics of the spacecraft are faster than the measurement rate. Moreover, the results obtained could heavily depend on the initial conditions given to the filter. Furthermore, with only 10–12 data packages per pass, the convergence of the Kalman filter cannot be guaranteed for that small amount of data.

Recently, a new and interesting approach to the determination problem has been proposed, where the temperature and absorbed heat fluxes are used to determine the orientation of the spacecraft [46,47]. Other studies have also explored the idea of obtaining the attitude from thermal data, such as [48,49]. This approach is particularly relevant for the UPMSat-2 satellite, which is equipped with temperature sensors. Furthermore, Gadalla Mohamed et al. [50] have shown that the rotation of a moving spacecraft can significantly affect the temperature difference between its surfaces, making it possible to determine the spacecraft's rotation rate. Therefore, this approach holds the potential to distinguish between fast and slow rotation rates of the satellite.

Given the current state of the art and the specific characteristics of the satellite, this paper adopts a non-traditional approach to solve the determination problem. Specifically, the angular velocity of the satellite is determined by using the stroboscopic effect and a transient thermal analysis.

3.2. Spacecraft attitude determination with stroboscopic effect

In order to determine the attitude of a spacecraft, it is necessary to first understand the available independent measurements, their uncertainties, and their nature. On the UPMSat-2 satellite, there are multiple sensors that can be used for attitude determination: 3 magnetometers and 6 photodiodes that estimate the solar direction. Additionally, housekeeping data such as solar panel currents and temperature measurements are recorded and are also useful for the satellite attitude determination [51]. Once the data packages are received and correlated with the satellite clock and TLE data, the orbit position is calculated, and relevant environmental conditions are determined, such as the solar direction and Earth's magnetic field in their respective reference frames.

Traditional attitude determination methods like TRIAD or q-method can then be used to calculate the attitude using the sensor measurements and environmental conditions for that specific spacecraft position. However, for this mission, estimating the angular velocity of the satellite is challenging due to the similarity between the expected rotation frequency and the measurement frequency (one revolution per minute and one measurement per minute). This produces an stroboscopic effect in which the apparent angular velocity is not the true angular velocity, as seen in Fig. 4, where the results of the Sun vector calculated using the sun sensor and solar panels are plotted. The x and y components of the solar vector appear to change slowly, while the z -component remains almost constant. This behavior implies that the satellite seems to rotate around its z -axis very slowly, but this appearance is due to the stroboscopic effect as it can be clearly seen when the results are compared to the expected evolution of the Sun vector. The modeled Sun vector depicted in the graph is obtained by simulating the satellite's rotation at a rate of 0.102 rad/s with an initial phase of 155 degrees. It is worth emphasizing that the simulated Sun vector evolution closely aligns with the measured data, enhancing the probability that a stroboscopic effect is indeed in play. Therefore given the presented results, in order to deduce the true angular velocity, and assuming that the control law is working as intended, it is necessary to assume the presence of an stroboscopic effect in the observed angular velocity.

To begin, let us assume that the spacecraft is rotating around one of its principal axes while taking measurements of the environment, such as the direction of the Sun vector or magnetic field. During

short periods of time, each component of the measurements vector will exhibit a similar behavior to the one illustrated in Fig. 4, where two components vary, and the one parallel to the angular velocity remains constant. Let us consider one of the non-constant components and refer to its measurement as y . The evolution of this component can be approximated as a sinusoidal signal with an angular frequency ω_c , which is equal to the satellite's rotation rate, and a phase φ_0 , that represents the initial attitude of the spacecraft. Considering the sampling period of the signal, T_s , consecutive measurements of this vector component can be expressed mathematically as follows:

$$y_i = \sin(\omega_c t_i + \varphi_0) = \sin\left(\frac{2\pi}{T_c} T_s i + \varphi_0\right) \quad i \in \mathbb{N} \quad (18)$$

Here, T_c represents the period of the motion, while i is a natural number indicating the measurement number (1, 2, 3, ...), and it is used as a subscript to identify each of the different measurements taken from the periodic signal. Usually, the sample frequency is higher than the frequency of the phenomenon, implying that $T_s \ll T_c$. In such cases, there is no stroboscopic effect, and the frequency of the signal can be determined. However, in the case of UPMSat-2, the satellite rotation frequency is expected to be similar to the sampling rate, i.e. $T_s \sim T_c$. Considering that the sampling period is numerically related to the rotation period, we can express it as follows:

$$T_s = k_{sc} T_c \quad (19)$$

where k_{sc} can be defined as the ratio between the sampling period, T_s , and the motion period, T_c , and can be also expressed as:

$$k_{sc} = \frac{T_s}{T_c} = 1 - r; \quad (20)$$

$$r = 1 - k_{sc} = \frac{T_c - T_s}{T_c}$$

where the new variable, r , has been defined for convenience. The observed signal can be then written as follows:

$$y_i = \sin(2\pi k_{sc} i); \quad i \in \mathbb{N} \quad (21)$$

as φ_0 is an arbitrary initial condition, it can be set to zero without losing generality. Then, substituting k_{sc} into expression (21), and using the definition given by Eq. (20), the observed signal can be expressed as:

$$y_i = \sin(2\pi(1-r)i) = \sin(2\pi i) \cos(-2\pi r i) - \cos(2\pi i) \sin(2\pi r i) \\ = -\sin(2\pi r i) \quad (22)$$

From Eq. (22), it is observed that the motion seems to have an apparent frequency f_{ap} , with the perception of being moving in the reverse direction if $T_c > T_s$. An apparent cycle is completed after $1/r$ measurements at sample rate T_s . Then, the apparent period, T_{ap} can be calculated as:

$$T_{ap} = \left(\frac{1}{r} T_s\right) \quad (23)$$

Therefore, the apparent frequency, can be expressed in the following way:

$$f_{ap} = \frac{1}{T_{ap}} = \frac{1}{\frac{1}{r} T_s} = \frac{r}{k_{sc} T_c} = \frac{r}{1-r} f_c \quad (24)$$

where f_c is the true frequency of the signal. This expression establishes a relationship between the apparent frequency and the true frequency of the motion. By substituting the definition of r given by Eq. (20) into (24) one can determine the true frequency based on the sample and the apparent frequencies.

$$f_c = \frac{1-r}{r} f_{ap} = \frac{f_c/f_s}{1-f_c/f_s} f_{ap} = f_s - f_{ap} \quad (25)$$

which can be finally expressed as:

$$\omega_c = \frac{2\pi}{T_s} - \omega_{ap} \quad (26)$$

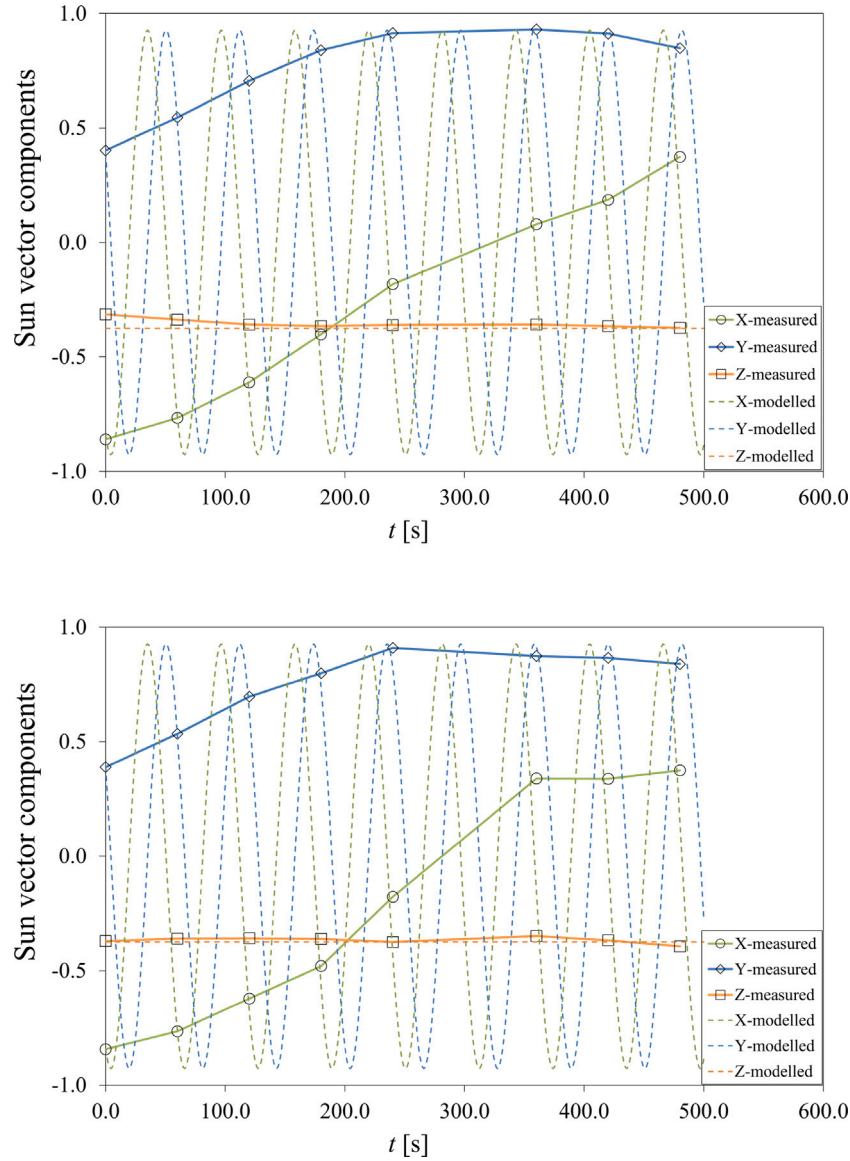


Fig. 4. Sun vector components regarding the UPMSat-2, calculated with the solar panels (a) and the Sun sensors (b), during one access to the ground station on September the 4th, 2020 compared to the expected evolution of the solar vector if the satellite rotates at 0.102 rad/s.

where ω_{ap} is the apparent angular velocity of the signal. With this expression, the angular velocity can be deduced if the sampling period and the apparent angular velocity are known. As the sampling period is fixed and known, the apparent angular velocity shall be calculated.

The Coriolis theorem describes how a magnitude in a moving or rotating frame is seen in the fixed frame. Mathematically, it is expressed as follows:

$$\dot{\mathbf{v}}_1^s = \dot{\mathbf{v}}_s^s + \boldsymbol{\omega}_{s1}^s \times \mathbf{v}_s^s \quad (27)$$

where $\mathbf{v}$ represents a generic vector. The subscripts indicate from which reference frame the variation is seen, and the superscript indicates the reference frame where the variation is expressed.

This theorem can be applied to obtain the angular velocity of a satellite based on any measurement taken in the satellite, taking into account that the environmental conditions are known in a fixed reference frame. In the case of UPMSat-2, the satellite is equipped with a sun sensor and three magnetometers. However, the magnetometers have a higher accuracy than the sun sensor, and their measurements can be combined into a single measurement. Consequently, the angular velocity of the satellite will be calculated using only the magnetometer

data. For the analysis, only the Earth's magnetic field will be considered, although the same approach could be applied to any other measurement that can be expressed as a direction in space. Therefore, by introducing the magnetic field into Eq. (27), it can be rewritten as follows:

$$\dot{\mathbf{B}}_s^s - \dot{\mathbf{B}}_1^s = \mathbf{B}_s^{s \times} \boldsymbol{\omega}_{s1}^s \quad (28)$$

where $\mathbf{B}_s^s$ represents the magnetic field measured by the magnetometer, while $\mathbf{B}_1^s$ represents the modeled magnetic field obtained through the IGRF magnetic model. Finally $\mathbf{B}^{\times}$ denotes the skew-symmetric matrix of vector $\mathbf{B}$.

$$\mathbf{B}^s = \begin{bmatrix} B_x^s \\ B_y^s \\ B_z^s \end{bmatrix} \quad \mathbf{B}^{s \times} = \begin{bmatrix} 0 & -B_z^s & B_y^s \\ B_z^s & 0 & -B_x^s \\ -B_y^s & B_x^s & 0 \end{bmatrix} \quad (29)$$

Then, $\dot{\mathbf{B}}_1^s$ and $\dot{\mathbf{B}}_s^s$ can be calculated as follows.

$$\dot{\mathbf{B}}_{1,i}^s = \frac{1}{T_s} \left[\mathbf{B}_{1,i}^s - \mathbf{B}_{1,i-1}^s \right] \quad (30)$$

$$\dot{\mathbf{B}}_{s,i}^s = \frac{1}{T_s} \left[\mathbf{B}_{s,i}^s - \mathbf{B}_{s,i-1}^s \right] \quad (31)$$

In this expression, i denotes the value of the magnitude at a certain instant. This implies that at least two consecutive magnetic field measurements are required. However, a significant issue with this method is that the skew-symmetric matrix is not invertible. Therefore, Eq. (28) does not possess a unique solution *a priori*. To address this issue, it is necessary to use the pseudo-inverse matrix to solve the system of equations [52]. The pseudo-inverse matrix is defined as follows:

$$\mathbf{W} = \left(\mathbf{B}_{s,i}^{s \times \times \top} \mathbf{B}_{s,i}^{s \times \times} \right)^{-1} \mathbf{B}_{s,i}^{s \times \times \top} \quad (32)$$

Finally, the apparent angular velocity is calculated as:

$$\omega_{s1,i}^s = \mathbf{W} \left[\dot{\mathbf{B}}_{s,i}^s - \dot{\mathbf{B}}_{1,i}^s \right] \quad (33)$$

It is possible to calculate the angular velocity of the spacecraft using only two consecutive measurements. However, this problem is ill-conditioned, which can lead to numerical difficulties. To avoid this issue, it is recommended to use three consecutive measurements instead.

$$\begin{bmatrix} \dot{\mathbf{B}}_{s,i-1}^s - \dot{\mathbf{B}}_{1,i-1}^s \\ \dot{\mathbf{B}}_{s,i}^s - \dot{\mathbf{B}}_{1,i}^s \end{bmatrix} = \mathbf{B}_{s,i}^{s \times \times} \omega_{s1,i}^s; \quad \mathbf{B}_{s,i}^{s \times \times} = \begin{bmatrix} \mathbf{B}_{s,i-1}^{s \times \times} \\ \mathbf{B}_{s,i}^{s \times \times} \end{bmatrix} \quad (34)$$

By solving the system of equations presented here and following the steps outlined in Eqs. (32) and (33), a unique and more precise solution can be obtained. When the sample frequency is twice the frequency of the motion, these equations provide the true angular velocity of the spacecraft. However, if the sample frequency is lower, it is only possible to determine the apparent angular velocity, and the stroboscopic effect must be considered. In such cases, the obtained results should be incorporated into Eq. (26) to determine the true angular velocity.

4. Results

4.1. Angular velocity results obtained for UPMSat-2

Fig. 5 shows the mean angular velocity obtained for the x - and y axes. The mean angular velocity shows a small component that is heavily influenced by the instruments noise and process errors. This suggests that the angular velocity of the x - and y axes is close to 0. Fig. 6 presents the results obtained for the mean angular velocity on the z -axis in each illuminated pass, once the stroboscopic effect is considered. The obtained angular velocity approximates to the target value of 0.1 rad/s.

As previously mentioned, once the satellite is stabilized using this control, the equilibrium angular velocity exhibits minor variations around the target velocity. A theoretical analysis was conducted using a target velocity of 0.1 rad/s and a control gain of $1 \cdot 10^8$, which revealed that the expected angular velocity varies between 0.1017–0.1033 rad/s along the orbit, as depicted in Fig. 3 [29]. As it can be observed in the results presented in Fig. 6, most part of the results are in the theoretical range and the maximum deviations are smaller than 3% from the bounds of the theoretical range.

Previous results are only valid if the satellite's angular velocity is similar to the sample rate and the stroboscopic effect is present. Therefore, it is necessary to validate or set lower and upper bounds to the angular velocity using other measurements that can indirectly and independently provide another value. To this end, thermal analysis can be used to determine the uniformity of temperature across the satellite's different faces and set a lower limit to the rotating speed [50]. Additionally, the method used to measure the magnetic field in the satellite allows for setting a preliminary upper limit on the rotation speed. The magnetometer readings are obtained as an average of five readings separated by 200 ms each. If the satellite rotates too fast, the resulting average will not have a physical meaning, and the control law cannot be applied. For instance, at one revolution per second, the average of the 5 readings would be zero, resulting in random values due to instrument noise. Even slower angular velocities can cause control problems. For example, at 30 revolutions per minute, the average value

of the readings is 90 degrees out of phase with the magnetic field at that moment. Therefore, one revolution per second (6.283 rad/s) seems like a reasonable upper limit for the angular velocity.

In order to study the expected thermal behavior of the UPMSat-2 at different spin rates, a reduced thermal model of the satellite was set-up based on the detailed thermal model created by IDR/UPM in ESATAN-TMS (see Fig. 7) [53]). Although the detailed thermal model contains much more information, its complexity makes unnecessarily complicated the calculation of the transient temperatures along the orbit (see work of Ref. [35] to observe the expected evolution of solar panel temperatures using a detailed thermal model). Therefore, a reduced model will be used. It consists of 7 nodes, 6 of them representing each panel of the satellite. The last node represents the inner part (trays, payloads, etc.) and it is conductively coupled with the external panels.

The parameters of the reduced model include thermal capacities, conductive couplings and the emissivity and absorptivity of the panels. A good initial estimation of the reduced model parameters can be obtained from the detailed model itself. Indeed, most of the parameters of the detailed thermal model were already correlated through the thermal vacuum testing results [53] and low uncertainty is expected in their values. However, there are some values that could not be correlated as, for example, those associated with the solar absorptivity of the external surfaces (due to lack of equipment for its measurement). Nevertheless, as the materials and surface treatments are known, it is possible to define upper and lower limits for the absorptivity.

Although the described initial thermal reduced model predicts the general thermal behavior of the UPMSat-2, a fine-tuning process is performed using the flight data received by the IDR/UPM's ground station during September 2020. A monotonic temperature increasing trend is observed despite of the flight data are distributed along many orbits. This is because the temperature data are plotted as a function of the time, starting at the eclipse exit instant, so the satellite is illuminated by the Sun (colored dots in Fig. 8).

The fine-tuning correlation can be performed on a case-by-case basis by estimating the initial condition, the spin rate, and the thermal parameters (thermal capacity, absorptivity, and emissivity) at the same time. The Sun vector and the angular velocity direction are also required, but they can be estimated through the onboard magnetometers and sun sensors [40]. Then, for each correlation, a better approximation between the data and the correlated model is obtained.

It should be taken into account that this correlation process is computationally expensive, so following this approach for every initial condition and spin rate is inefficient. Therefore it will be used to discard, before any further analysis, values of apparent angular velocities of around 0.0031 rad/s. In fact, trying to correlate a low spin rate model against the flight data, such as the one with apparent angular velocity of 0.0031 rad/s, yields to a thermal model whose parameters do not have a physical meaning. That is, the thermal capacity, the absorptivity and the emissivity of the panels acquire values outside the expected ranges.

Another approach for the fine-tuning correlation is to assume an infinite spin rate. By doing so, the correlation is simpler (the initial conditions do not have an effect and the angular velocity is known). Then, the thermal parameters can be adjusted so the model predicts the global averaged temperature trend. With this approach, the thermal parameters can be set, and a subsequent optimization process can be followed to find the spin rate that better match the data. This assumption works well if the temperature oscillation due to the spin is small compared to the secular variation. By looking at the temperature data of each individual pass (for example the pass depicted in Figs 9 and 10), it is possible to see that temperature oscillation around the global trend is small (around 1 °C or less). Therefore, the high-spin rate model should work well in this case.

After the correlation, the thermal model obtained (see solid curve in Fig. 8) predicts the evolution of the satellite average temperatures. The maximum discrepancy between the data and the high spin rate model is around ± 3 °C. There are mainly two possible sources for

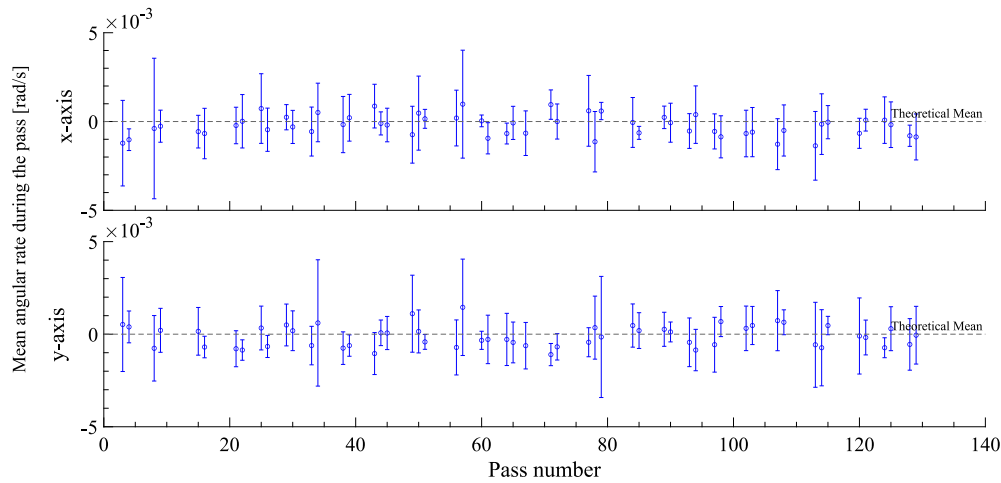


Fig. 5. UPMSat-2 mean angular velocity results for the x -axis and y -axis for each pass above the ground station. (In these results, only those passes that were illuminated and have more than five consecutive data points were considered).

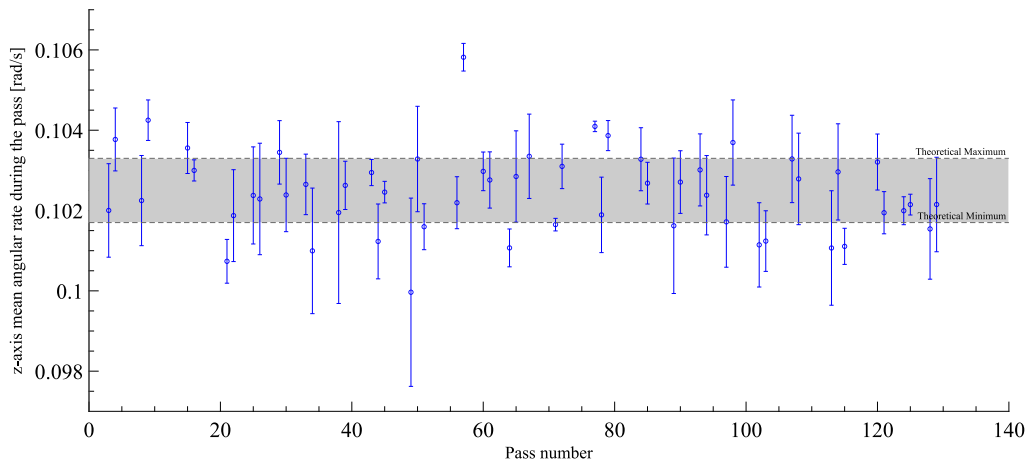


Fig. 6. UPMSat-2 mean angular velocity results for the z -axis for each pass above the ground station.

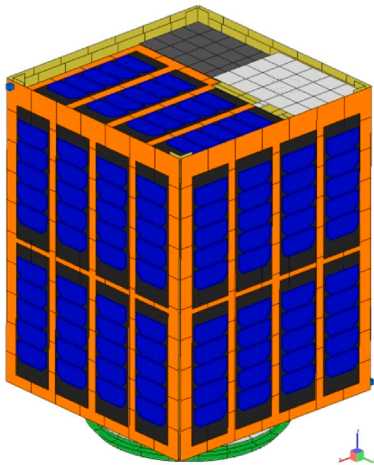


Fig. 7. UPMSat-2 detailed Geometrical Mathematical Model (GMM).

that discrepancy: a smaller spin rate that generates some oscillation around the average temperature, or a different initial temperature condition for each orbit. As the oscillation around the global trend is small, the $\pm 3^\circ\text{C}$ discrepancy is mainly due to differences in the initial condition, which in the end can be explained by small differences in the

operational mode, such as the solar panels working point or the internal dissipation of the electronics and the heaters, or differences that are propagated for several orbits, but cannot be modeled with the available information. In Fig. 8, the red dashed line represents the correlated model, simulated at an apparent angular velocity of 0.0031 rad/s. The plot clearly shows why 0.0031 rad/s or other low rotation spins are not compatible with the data. Low angular velocities create relatively extended periods without sun illumination at the faces, resulting in a temperature decrease out of eclipse that the data does not exhibit.

Summarizing, the general trend of the thermal data received in each pass can be reproduced in the model assuming an adequate initial temperature and a fast spin of the satellite. In addition, some cyclical variations are observed, and those should be explained by adjusting the model to the exact rate of rotation of the satellite in its orbit.

Once the thermal model is set-up, it is possible to start searching for the spin rate that better fit the data, but each pass needs to be considered individually. This process is necessary because, as stated before, the temperature variations with respect to the global trend produced by the spin are considerably smaller than the variations produced by the different initial conditions. Therefore, the contribution of the initial conditions needs to be set individually in each pass, in order to clearly see the oscillations due to the spin.

The approach followed here is to optimize, for each pass, the initial temperature and the initial attitude. Because the Sun rays and the spin direction are known, the attitude is specified only by setting the initial rotation angle (between 0 and 2π). The initial attitude determines the

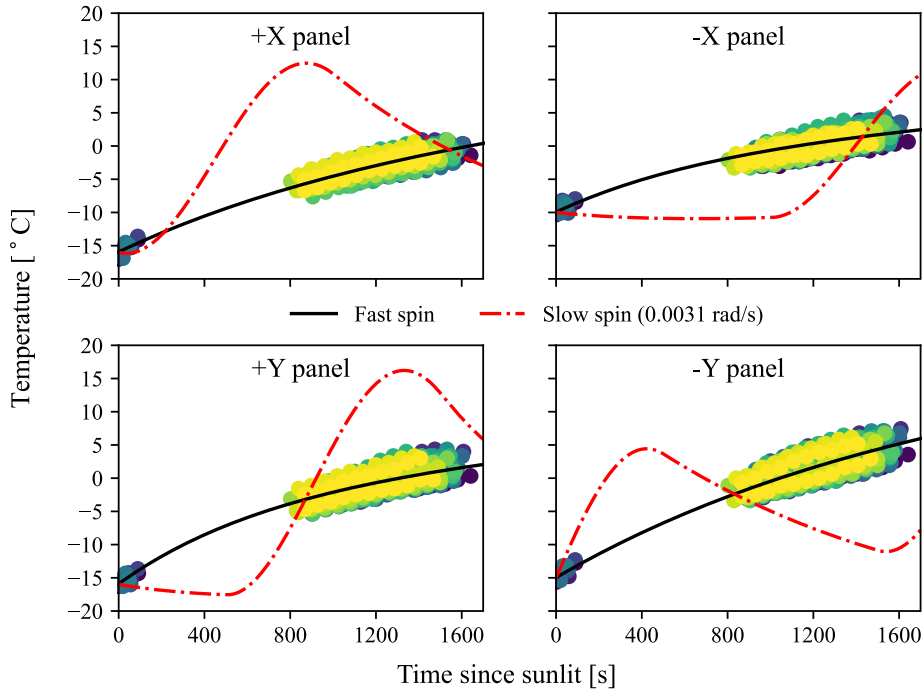


Fig. 8. Flight temperature data vs time starting at the satellite eclipse exit for several passes (colored points) and correlated model assuming fast (black solid line) and slow (red dashed line) spin rates.

phase of the temperature oscillations, and its determination becomes more important the lower the spin rate is. The initial temperature range considered is the average shown in Fig. 8 with ± 5 °C. Once a spin rate is set, the optimization begins with two free parameters, the initial temperature, and the rotation angle. The initial conditions have been calculated as those that minimize the root of the residual sum of the squares between the data and the model. For this purpose, a non-linear least squares minimization algorithm based on quasi-Newton steps was used.

In Fig. 9, the result after the optimization process (green solid line) is shown for one of the passes at an spin rate of 0.03 rad/s. The red and blue dotted lines, labeled as Hot IC and Cold IC respectively, represent a model without optimum initial conditions. The optimization process starts with some arbitrary initial conditions, for example the Hot IC or the Cold IC, and iteratively converge to other initial conditions (Optimum IC) that better fits the data.

In Fig. 10 the same flight data as in Fig. 9 is presented, together with the correlated model with optimized initial conditions, and for four different spin rates. It is possible to note that the flight data have in fact small oscillations that high spin rates (0.2 rad/s or higher) cannot model. This can also be stated for low spin rates, of around 0.03 rad/s or lower (see also solid green line of Fig. 9 and slow spin curve of Fig. 8) as the amplitude of their oscillations in the model are too high and do not correspond to the flight data. If this analysis is done for other passes, the plots are similar and same conclusions can be extracted. For spin rates in between, it is not easy to see directly which one fits better with the data, but it is possible to numerically compare the goodness of fit by defining the following deviation between the model and the data:

$$\varepsilon_T = \sqrt{\frac{\sum_{i=1}^{N_p} \sum_{j=1}^{N_d} (T_{m,ij} - T_{data,ij})^2}{N - 1}} \quad (35)$$

where N is the total amount of temperature data, which is the sum of the N_d temperature points available on each pass during a total of N_p passes. $T_{data,ij}$ is the flight data j during the pass i and corresponds to the model temperature $T_{m,ij}$. For this analysis a total of 4004 temperature data points were used. In Fig. 11 the temperature deviation,

ε_T , between the model and the flight data is plotted for different spin rates. As expected, when the spin rate increases, the error decreases reaching a stationary value for high spin rates. There are, however, three relevant local minimum points in the plot: 0.052 rad/s, 0.10 rad/s and 0.21 rad/s. This is not a coincidence, but those values are multiple of the most likely satellite spin rate. However, the global minimum is around 1 rpm and is the most plausible spin rate for the UPMSat-2. The local minimum at 0.052 rad/s seems to be caused because the simulated spin is half the spin rate. However, the differences between maximum and minimum in the model oscillations (see the dotted blue curve in Fig. 11) are higher than those in the data. Something similar but opposite can be said for the 0.21 rad/s spin rate, which doubles the spin rate, whose oscillations are smaller than those observed in the data.

The foregoing shows that an angular velocity speed in the vicinity of 0.10 rad/s is the most likely for the satellite and coincides with the results obtained in Fig. 6 assuming the stroboscopic effect of the measurement rate. With this speed of rotation and adequate initial conditions, the thermal model of the satellite can be realistically adjusted to the thermal data of the passes both in global trend and in amplitude of the cyclical component. On the other hand, if the apparent angular speed is taken as good, the results of the attitude control could be adjusted but the thermal results would not be consistent.

4.2. Results regarding the orientation of the UPMSat-2

To evaluate the performance of the control law, it is necessary to analyze the error between the actual satellite's attitude and the expected one. The UPMSat-2 control law is designed to stabilize the z -axis almost perpendicular to the orbital plane, with the $-z$ face pointing in the positive direction of the normal to the orbit. Since the satellite is in a 10:30 Sun-synchronous orbit, the incidence angle of the Sun with the normal to the orbital plane remains nearly constant throughout the month of observation, with a mean value of 110.7 degrees for the month of September. Therefore, the incidence angle of the Sun with respect to the normal to the $-z$ face of the satellite serves as a good indicator of the pointing precision of the control law.

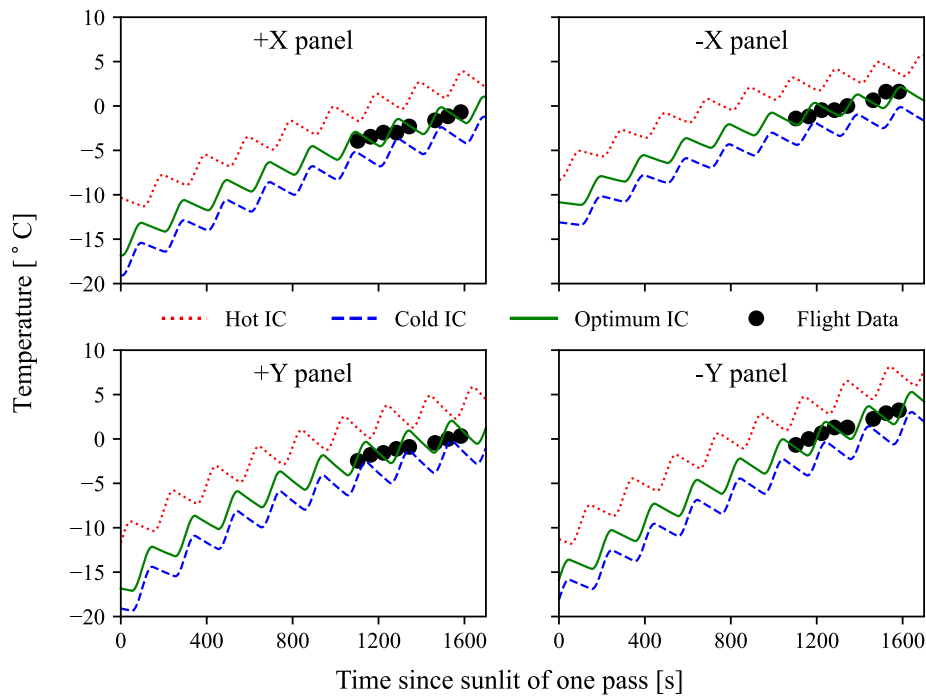


Fig. 9. Optimization of the initial conditions (IC) for each orbit and each spin rate. The plot corresponds to a model with a spin rate of 0.03 rad/s.

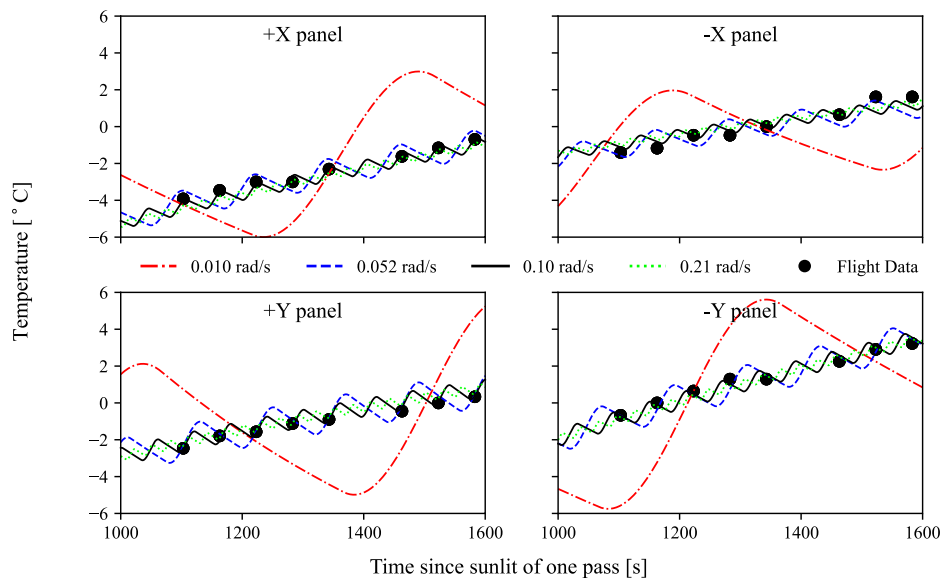


Fig. 10. Model temperatures for different spin rates (lines) and flight temperature measurements (dots) for one pass.

To assess the pointing precision, an histogram of the Sun incidence angle with the z -axis (referred to as the illumination angle from now on) is plotted in Fig. 12. The distribution of the incidence angle can be fitted to a normal distribution with a mean of 112.9 degrees and a standard deviation of 2.5 degrees. The figure also shows the mean theoretical illumination angle if the z -axis were perfectly aligned with the normal direction of the orbital plane (represented by the vertical dashed line). On average, there is a difference of 2.2 degrees between the normal direction of the orbit and the pointing direction of the z -axis. This behavior is in good agreement with the expected misalignment for the inclination of the orbit, as shown in Fig. 2 [29].

5. Conclusions

In this work, the in-orbit performance of the UPMSat-2 attitude control law was studied. The magnetic control law governing the UPMSat-2 attitude is derived from the B-dot control law, including an additional performance: it can stabilize an axis in an orientation almost perpendicular to the orbital plane, and then control the angular velocity around this axis to the desired target velocity.

The analysis of the data from the satellite has been challenging due to the low sampling rate (one data package per minute), being similar to the expected period of the satellite's main motion (around one revolution per minute). This fact causes an stroboscopic effect and

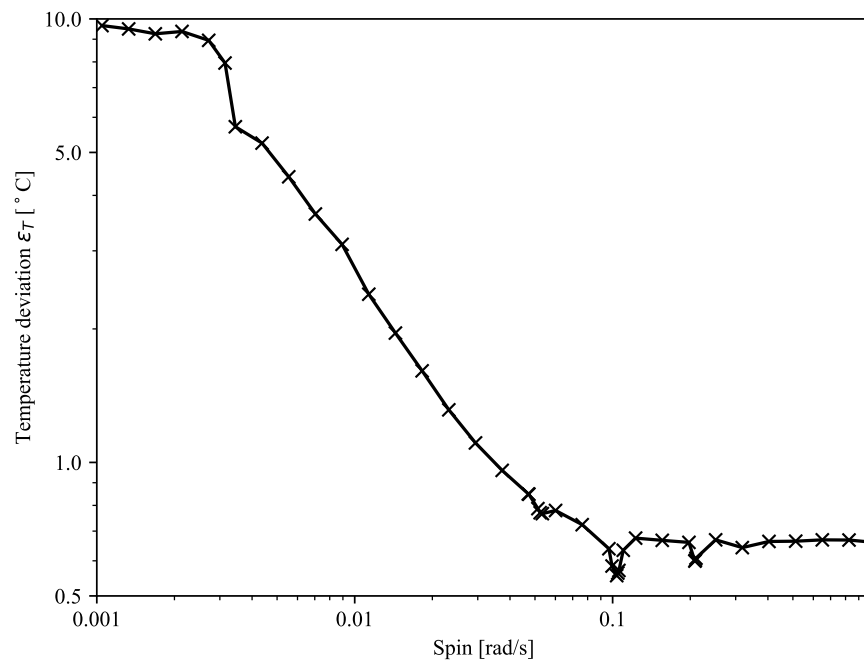


Fig. 11. Variation of the temperature deviation between the flight data and the model as a function of the spin rate.

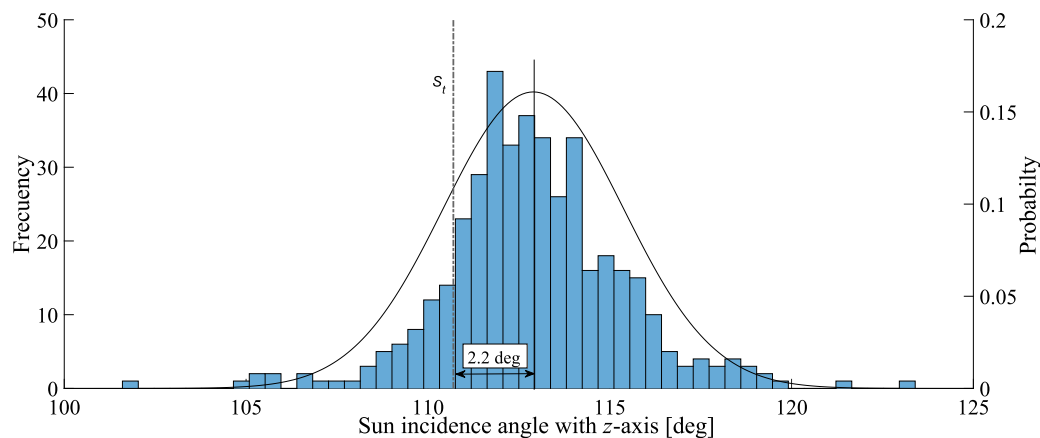


Fig. 12. Assessment of the pointing control of the UPMSat-2. Histogram of the z-axis illumination angle compared to the mean value, S_t , of the theoretical illumination angle.

therefore the reconstruction of the original signal is required to analyze the satellite in-orbit attitude.

After correcting that stroboscopic effect, the magnetometer data have shown good agreement with the expected results. In addition, to verify that the satellite is not rotating at its apparent angular velocity, a thermal analysis has been conducted. The thermal analysis ruled out rotation rates under 0.052 rad/s, as they did not match the general temperature trend of the satellite. The analysis has also shown that only high rotation rates could fit with the experimental data, and an spin rate near 0.1 rad/s has the best agreement.

Additionally, the control law has been evaluated using Sun sensors to determine the alignment of the z-axis with the normal direction to the orbital plane. The results demonstrate that the control law maintain a close alignment with a mean deviation of 2.2 degrees, which is consistent with the theoretical deviation range.

CRedit authorship contribution statement

Angel Porras-Hermoso: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing

– original draft, Writing – review & editing. **Javier Piqueras:** Formal analysis, Investigation, Methodology, Software, Visualization, Validation, Writing – original draft. **Javier Cubas:** Conceptualization, Formal analysis, Investigation, Methodology, Supervision, Validation, Writing – original draft, Writing – review & editing. **Elena Roibás-Millán:** Data curation, Project administration, Resources, Supervision, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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